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A Convex Optimization Approach to Curve Fitting with B-Splines

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Abstract: This paper presents a framework for approximating data with smooth splines. The classical spline approximation problem is reformulated as a convex optimization problem, in which both the required number of knots and the knot locations are found automatically and simultaneously. Spline constraints are easily added to improve the quality of the approximation. Three examples are presented to illustrate the effectiveness of the proposed framework. The obtained numerical results show improvements of the smoothness of two benchmark problems and show that more complex constraints can be included.

Keywords: convex optimization, filtering, smoothing

1. INTRODUCTION

Approximating measured data by a smooth curve is a frequent problem in motion analysis, computer aided design, image processing and many other fields. When the data exhibit a complicated shape, simple polynomials fall short. In this case, piecewise polynomials or *polynomial splines* are often chosen as parametrization for the underlying function [Dierckx, 1993, de Boor, 2001].

The main challenge when optimizing splines is determining the locations of the breakpoints or *knots*. This requires treating the knots as variables, resulting in a highly nonlinear and nonconvex optimization problem [Dierckx, 1993]. Consequently, it is difficult to obtain and guarantee global optimality.

In the literature many methods have been proposed to solve this problem. Most methods, however, require a good initial guess of the knot locations [Jupp, 1978], cannot guarantee global optimality [Hayes, 1974, Dierckx, 1993, Molinari et al., 2004] or produce redundant knots [Dierckx, 1993]. Also, one often wishes to incorporate knowledge of the underlying function in the optimization. Dierckx [1993] proposes algorithms to deal with simple constraints such as end-point derivatives, periodicity and convexity, but more general constraints cannot be incorporated.

Demeulenaere et al. [2009] propose a convex framework for optimizing rigid motion trajectories with splines. In their work, the spline knots are optimized indirectly by supplying many candidate knot locations and using a regularization to favor solutions with few active knots. This approach results in a convex optimization problem for which the global optimum is guaranteed to be found efficiently and reliably. In this paper we apply this framework to the curve fitting problem with following essential modifications: (i) instead of integrating a piecewise linear function up to the spline degree, B-splines are used as a basis to evaluate polynomial splines, which enhances numerical stability and provides an elegant way to enforce semi-infinite constraints without gridding, and (2) to further increase the sparsity of the solution a reweighted ℓ_1 minimization [Candès et al., 2008] is added to the framework. Also, knowledge of the underlying function is easily incorporated by adding (convex) constraints to the optimization problem.

This paper is organized as follows. Section 2 introduces polynomial splines and the convex programming framework for spline approximation. In Section 3 examples, with and without constraints, are presented to illustrate the effectiveness of the developed framework.

2. BASIC OPTIMIZATION PROBLEM

2.1 Polynomial Spline Basics

A function $s(x)$ defined on a finite interval $[x_{\min}, x_{\max}]$ is a polynomial spline of degree $k \geq 0$, having as knots the strictly [1] increasing sequence λ_i, $i = 0, 1, \ldots, g+1$ if [Dierckx, 1993]:

(1) $s(x)$ is a polynomial of degree $\leq k$ on each knot interval $[\lambda_i, \lambda_{i+1}]$:

$$s_{[\lambda_i,\lambda_{i+1}]} \in \mathcal{P}_k, i = 0, 1, \ldots, g\,. \tag{1}$$

(2) $s(x)$ and its derivatives up to order $k-1$ are continuous on $[x_{\min}, x_{\max}]$:

$$s(x) \in \mathcal{C}^{k-1}_{[x_{\min},x_{\max}]}\,. \tag{2}$$

It is assumed that $\lambda_0 = x_{\min}$ and $\lambda_{g+1} = x_{\max}$. The knots λ_i, $i = 1, \ldots, g$ are called the internal knots. An internal knot λ_i is called active if the kth-order derivative of $s(x)$ is discontinuous at λ_i and is inactive otherwise. Fig. 1 shows a cubic ($k = 3$) spline with 6 internal knots and its first three derivatives. The spline consists of polynomial segments of at most degree 3 and, as allowed by (2), features discontinuous jumps in $s^{(3)}(x)$ at 4 active knots.

[1] If the knot sequence is not strictly increasing the continuity condition (2) must be relaxed.

In this paper, B-splines are used as a basis to represent polynomial splines. A B-spline basis function of degree l (order $l+1$), defined on $[\lambda_i, \lambda_{i+l+1}]$, is computed using the Cox-de Boor recursive formula [de Boor, 2001]

$$B_{i,l+1}(x) = \frac{x-\lambda_i}{\lambda_{i+l}-\lambda_i} B_{i,l}(x) + \frac{\lambda_{i+l+1}-x}{\lambda_{i+l+1}-\lambda_{i+1}} B_{i+1,l}(x)\,, \tag{3}$$

starting with

$$B_{i,1}(x) = \begin{cases} 1, \text{if } x \in [\lambda_i, \lambda_{i+1})\,, \\ 0, \text{if } x \notin [\lambda_i, \lambda_{i+1})\,. \end{cases} \tag{4}$$

Note that $B_{i,l}(x) \geq 0, \forall x$. This property will be used in section 3.2.

For a given knot sequence $\lambda_0, \ldots, \lambda_{g+1}$, the dimension of the vector space of functions satisfying (1) and (2) is $g+k+1$ [Dierckx, 1993]. Yet, with the given set of knots, we can only construct $g-k+1$ linearly independent B-splines of degree k. To obtain a full set of basis functions, we need an additional $2k$ independent B-splines. By introducing the boundary knots $\lambda_{-k}, \ldots, \lambda_{-1}$ and $\lambda_{g+2}, \ldots, \lambda_{g+k+1}$ satisfying

$$\lambda_{-k} = \ldots = \lambda_{-1} = \lambda_0 = x_{\min}\,, \tag{5}$$

$$\lambda_{g+1} = \lambda_{g+2} = \ldots = \lambda_{g+k+1} = x_{\max}\,, \tag{6}$$

$B_{i,k+1}(x)$ can be calculated for $i = -k, \ldots, -1$ and $i = g-k+1, \ldots, g$. This choice of boundary knots implies that all B-splines vanish outside $[x_{\min}, x_{\max}]$. Every spline $s(x)$ now has a unique representation

$$s(x) = \sum_{i=-k}^{g} c_i B_{i,k+1}(x)\,, \tag{7}$$

where c_i are called the B-spline coefficients of $s(x)$.

The νth derivative of $s(x)$ is itself a spline of degree $k-\nu$

$$s^{(\nu)}(x) = \prod_{i=1}^{\nu}(k+1-i) \sum_{i=-k+\nu}^{g} c_i^{(\nu)} B_{i,k+1-\nu}(x)\,, \tag{8}$$

with

$$c_j^{(i)} = \begin{cases} c_j\,, & \text{if } i = 0\,, \\ \frac{c_j^{(i-1)} - c_{j-1}^{(i-1)}}{\lambda_{j+k+1-i} - \lambda_j} & \text{if } i > 0\,. \end{cases} \tag{9}$$

2.2 Polynomial Spline Optimization

The basic approximation problem that is considered can be formulated as follows [Hayes, 1974]: given values y_r, $r = 1, \ldots, m$, corresponding to values $x_r \in [x_{\min}, x_{\max}]$, determine a function $y(x) := y(x;\theta)$ of known form but containing a vector θ of n disposable parameters to be determined such that $y(x_r) \approx y_r$. Also, it is desirable to be able to include knowledge of the underlying function to improve the quality of fit.

The classical way to solve this approximation problem with polynomial splines, is to treat both the spline coefficients c_i, $i = 1, \ldots, g+k+1$, and the spline knots λ_j, $j = 1, \ldots, g$ as disposable parameters. However, this approach makes the spline approximation problem nonconvex, because the spline $s(x)$ is a nonconvex function of the knots. To overcome the nonconvexity we here adopt an indirect spline knot optimization approach proposed by Demeulenaere et al. [2009]. Many (typically 500 to 1000) fixed candidate knot locations are provided, leaving only

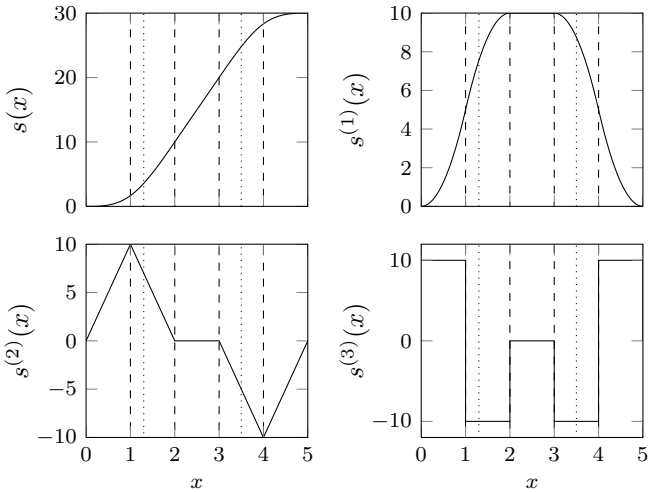

Fig. 1. A cubic splines and its first three derivatives. The dashed vertical lines indicate the active knots and the dotted lines denote the inactive (redundant) knots.

the spline coefficients to be determined and in order to favor solutions with few active knots an ℓ_1 regularization is used. This strategy results in a convex problem of which the global optimum is guaranteed to be found efficiently and reliably using dedicated algorithms.

Given k and the knot sequence $x_{\min} = \lambda_{-k} = \cdots = \lambda_0 \leq \cdots \leq \lambda_{g+1} = \cdots = \lambda_{g+k+1} = x_{\max}$, the spline is completely determined by its spline coefficients c_i, which are taken as the optimization variables. To calculate the derivatives easily, their spline coefficients are also added as variables. The coefficients are arranged in the optimization vector $\mathbf{c} \in \mathbb{R}^{(k+1)(g+1+\frac{k}{2})}$ as

$$\mathbf{c} = [\underbrace{c_{-k}^{(0)} \ldots c_g^{(0)}}_{g+k+1} \underbrace{c_{-k+1}^{(1)} \ldots c_g^{(1)}}_{g+k} \ldots \underbrace{c_0^{(k)} \ldots c_g^{(k)}}_{g+1}]^T\,. \tag{10}$$

Since there are only $g+k+1$ independent variables, $k(g+\frac{k+1}{2})$ constraints must be added. These constraints are found in equations (9), which are simple linear equality constraints that are added to the problem as $C \cdot \mathbf{c} = 0$ where C is a sparse banded matrix.

For a spline with knots λ_i and degree k and for measurements (x_r, y_r), the basic approximation problem we want to solve is

$$\underset{\mathbf{c}}{\text{minimize}} \sum_{i=1}^{g} w_i \left| s^{(k)}(\lambda_i+) - s^{(k)}(\lambda_i-) \right| \tag{11a}$$

$$\text{subject to} \sum_{r=1}^{m} (v_r(y_r - s(x_r)))^2 \leq S \tag{11b}$$

$$C \cdot \mathbf{c} = 0 \tag{11c}$$

$$g(s(x), \ldots, s^{(k)}(x)) = 0 \tag{11d}$$

$$h(s(x), \ldots, s^{(k)}(x)) \leq 0, \tag{11e}$$

with w_i and v_r chosen weights, S a fixed parameter which controls the quality of fit, and $s(x)$ and $s^{(i)}(x)$ as defined by equations (7) and (8) respectively. The objective function (11a) is a measure of the non-smoothness of the fit. $g(s(x), \ldots, s^{(k)}(x))$ and $h(s(x), \ldots, s^{(k)}(x))$ are additional constraints to the problem, e.g. prior knowledge

of the underlying function. If g is linear and h is convex, this minimization problem is convex and, although it has thousands of variables, it can be solved in under a second on a regular PC.

Problem (11) is similar to the so-called smoothing criterion for approximation [Dierckx, 1975]. Differences are found in the following:

- Instead of using the ℓ_1-norm in equation (11a), Dierckx [1975] uses the ℓ_2-norm

$$\sum_{i=1}^{g} \left(s^{(k)}(\lambda_i+) - s^{(k)}(\lambda_i-) \right)^2 . \tag{12}$$

  Where the ℓ_2-norm only smooths the fit, the ℓ_1-norm has a twofold effect:
  (1) The ℓ_1-norm (11a) measures the non-smoothness of $s(x)$.
  (2) It is well-known in the area of function approximation that ℓ_1-norm minimization is likely to yield sparse solutions. Applied to the present problem, sparsity implies few nonzero elements in $(s^{(k)}(\lambda_i+) - s^{(k)}(\lambda_i-))$, that is, few jumps in $s^{(k)}(x)$ or in other words few active knots. This means the optimal number of knots and optimal knot locations are found automatically, whereas in other algorithms the number of knots must be chosen beforehand and often a good initial guess of the knot locations is required for the algorithm to converge.
- Since we solve a convex problem, arbitrary convex constraints (11d) and (11e) can be included without making the problem substantially more difficult. In Dierckx [1993] every other constraint requires a change in algorithm and therefore only algorithms exist for few different types of constraints (e.g. end-point derivative constraints and linear inequality constraints).

If still too many active knots are observed in the solution, a reweighted ℓ_1 minimization is performed to improve sparsity of the solution [Candès et al., 2008]. Initially $w_i = 1$, $i = 1, \ldots, g$. In each reweighting iteration j, the ℓ_1 optimization problem (11) is solved and the weights w_i are updated as:

$$w_i^{(j+1)} = \frac{1}{\left| s^{(k)}(\lambda_i+) - s^{(k)}(\lambda_i-) \right| + \varepsilon} , \tag{13}$$

where ε should be chosen slightly smaller than the expected nonzero magnitudes of $s^{(k)}(\lambda_i+) - s^{(k)}(\lambda_i-)$.

3. EXAMPLES

Three examples illustrate the effectiveness of the developed framework. A first example illustrates the framework without additional constraints for well-known data that are often used to test spline fitting algorithms. A subsequent example where concave stress-strain data are fitted, shows that constraints can easily be added to the optimization problem. A final example illustrates data fitting for mutually constrained data sets. All examples are solved with MATLAB® using `cvx`, a package for specifying and solving convex programs [Grant and Boyd, 2010], with SDPT3 [Toh et al., 1999] set as solver.

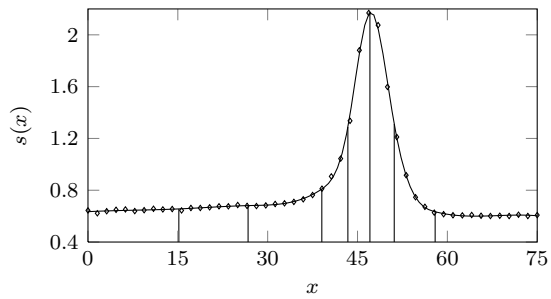

Fig. 2. A cubic spline ($k = 3$) fitted to the titanium heat data from de Boor and Rice [1968]. The residual quadratic error is $S = 0.0073$. The vertical lines represent the locations of the active knots.

method	g	non-smoothness
Dierckx [1975]	8	0.0899
Jupp [1978]	5	0.0742
Convex approach	7	0.0433

Table 1. Comparison with respect to the degree of non-smoothness of different fits to the titanium heat data

3.1 An unconstrained example

As a first example the titanium heat data from de Boor and Rice [1968] are fitted. The data are known to be difficult to fit using traditional techniques due to the sharp peak in the data (Fig. 2), and have therefore often been used to test spline fitting algorithms [de Boor and Rice, 1968, Jupp, 1978, Dierckx, 1993, Yoshimoto et al., 2003, Molinari et al., 2004]. The basic optimization problem (11) is solved for a cubic spline ($k = 3$) with $S = 0.0073$, $v_r = 1$ and 1000 equidistant candidate knots. The value for S is close to the theoretical minimum quadratic error for 5 knots [Jupp, 1978] and corresponds to the value used in Dierckx [1975] so a comparison can be made. Fig. 2 shows the fit and the active knot locations found with our method.

Seven active knots are found after seven reweighting iterations (13) with $\varepsilon = 10$. Because we solve a convex problem, the global optimum is attained for the supplied knots. This is confirmed when comparing smoothness values with Jupp [1978], Dierckx [1975]. In Table 1, the degree of non-smoothness, as in (12) is shown for our solution and these of Jupp [1978] and Dierckx [1975]. The attained smoothness of our fit is clearly better. Moreover, our method neither requires an initial guess of the knot sequence, nor is it necessary to choose the number of knots.

3.2 Convexity Constraints

If there is any knowledge of the underlying function, such as shape properties, periodicity and known function values, it should be included in the optimization problem to improve the quality of fit. In this example we consider stress-strain data from Amos and Slater [1969], of which the underlying function is known to be concave.

Shape preserving conditions like positivity, monotonicity and convexity are easily derived from the positivity property of B-splines. Since $B_{i,l}(x) \geq 0, \forall x$, $s(x)$ is positive if

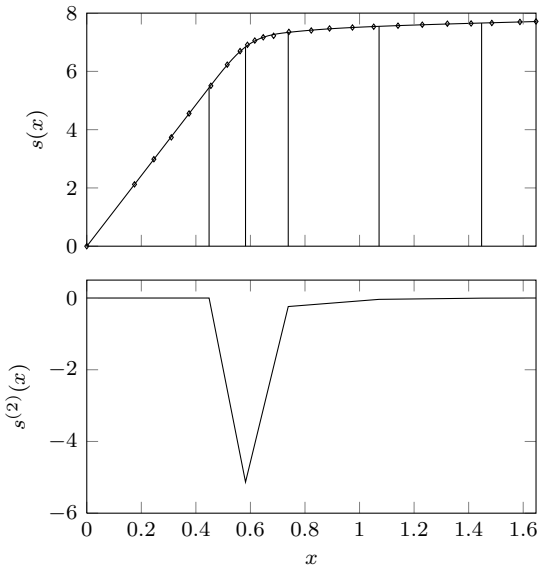

Fig. 3. A cubic spline (top) and its second derivative (bottom) fitted to the concave stress-strain data from Amos and Slater [1969], with a residual quadratic error $S = 0.0044$. The concavity constraint is active at the first and final segment. The vertical lines represent the locations of the active knots.

$c^{(0)} \geq 0$, monotonically increasing if $c^{(1)} \geq 0$ and convex if $c^{(2)} \geq 0$. These conditions, however, are only sufficient and may introduce conservatism [de Boor and Daniel, 1974]. The constraint $c^{(2)} \leq 0$ is added to the optimization problem (11) to impose concavity, and the problem is solved for a cubic spline with $S = 0.0044$, as in Dierckx [1980], and 1000 equidistant candidate knots.

Five knots are found after five reweighting iterations (13) with $\varepsilon = 10$. Fig. 3 shows the fit (top) and its second derivative (bottom). The concavity constraint is active on the first and last polynomial segment, resulting in linear segments in $s(x)$.

Both the convex approach and Dierckx [1980] find five active knots, although the latter only allows knots at the measurement points. The convex framework yields a smoothness value of 500 whereas Dierckx [1980] finds 828.

3.3 Fitting with Constraints Between Curves

In this section more complex constraints are added that, to the best of the authors' knowledge, cannot be included with other spline fitting algorithms. We consider measurements of n marker positions P_i, $i = 1, \ldots, n$, at time samples t_k, $k = 1, \ldots, m$, on a rigid body moving in 3D space as illustrated in Fig. 4 for $n = 3$. Through measured coordinates, x_i, y_i and z_i of each marker, we wish to fit a spline, s_{x_i}, s_{y_i} and s_{z_i}, taking into account that the distance between two markers remains constant. Yet, due to measurement noise, imposing a constant distance would yield a bad quality of fit. Therefore we allow the distance to vary $\pm\delta$ around the true distance μ_{ij} between the markers P_i and P_j $(i < j)$, for which an approximation is found by averaging out the distance between the measurements of

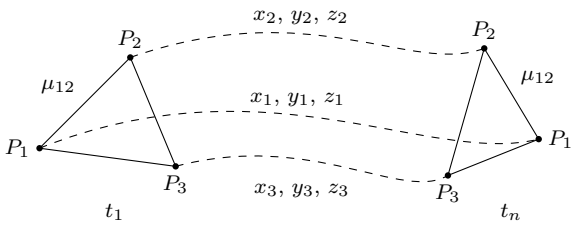

Fig. 4. Three markers, P_1, P_2 and P_3 measure the movement of a rigid body. At all times the distance between the markers remains the same.

the marker positions over time. Then, the constraints we wish to impose are

$$(\mu_{ij} - \delta)^2 \leq \sum_{l=x,y,z} (s_{l_i} - s_{l_j})^2 \leq (\mu_{ij} + \delta)^2, \tag{14}$$

with $i < j$. In order to be able to impose this semi-infinite constraint, it is considered only at the time samples t_k. However, adding this constraint to (11) would render the problem nonconvex, due to the concavity of the lower bound. To overcome this problem, the lower bound in (14) is linearized.

For simplicity, Fig. 5 illustrates the linearization in two dimensions (constant z-coordinate). In the $(\Delta x, \Delta y) = (x_i - x_j, y_i - y_j)$-plane, constraint (14) is visualized by two concentric circles with radii $(\mu_{ij} \pm \delta)$. Now suppose we have from the measurements a point $(dx, dy) = (x_i(t_k) - x_j(t_k), y_i(t_k) - y_j(t_k))$. Because of (11b), $(s_{x_i}(t_k) - s_{x_j}(t_k), s_{y_i}(t_k) - s_{y_j}(t_k))$ will lie close to this point. Therefore the concave lower bound is linearized at a point (n_x, n_y) on the lower bound that is closest to (dx, dy),

$$\begin{aligned} n_x &= \operatorname{sgn}(dx)(\mu_{ij} - \delta)\sqrt{\frac{dx^2}{dx^2 + dy^2}}\,, \\ n_y &= \operatorname{sgn}(dy)(\mu_{ij} - \delta)\sqrt{\frac{dy^2}{dx^2 + dy^2}}\,, \end{aligned} \tag{15}$$

resulting in the convex set C, indicated by the gray region in Fig. 5, defined by the constraints

$$(s_{x_i} - s_{x_j})^2 + (s_{y_i} - s_{y_j})^2 \leq (\mu_{ij} + \delta)^2\,, \tag{16}$$

$$n_x(s_{x_i} - s_{x_j}) + n_y(s_{y_i} - s_{y_j}) \geq (\mu_{ij} - \delta)^2\,. \tag{17}$$

Notice the use of the sign-function in (15) to ensure that the linearization is carried out in the correct quadrant.

To verify the potential of the above approach, fictitious measurements are generated for two markers on a rigid body, moving in a plane. The markers are initially located at $(1\,\mathrm{m}, 0\,\mathrm{m})$ and $(-1\,\mathrm{m}, 0\,\mathrm{m})$ and rotate 180° around the origin in $N = 100$ samples with $t_s = 0.01\,\mathrm{s}$. Gaussian noise with a standard deviation $\sigma = 0.02\,\mathrm{m}$ is added to each coordinate. Fig. 6 illustrates the simulated marker measurements and the true trajectories for both markers.

Four splines are then fitted simultaneously; one for each coordinate of each marker. Problem (11) is therefore extended.First, for each coordinate a one-norm regularization is added to the goal function and constraints (11b, 11c) are repeated. Furthermore, constraints (16, 17) are added to constrain the distance between the markers. For the optimization S is chosen equal to $N\sigma^2$, since this is the expected value of the residual quadratic error, $\delta = 0.001\,\mathrm{m}$, $k = 4$, $g = 100$, and no reweighting iterations are performed.

The resulting splines all have no or one active knot. The fits with no active knots are similar to higher order taylor expansions of the trajectory.

Fig. 7 shows the resulting distance in function of time for fitted splines with and without the distance constraints. The dotted line indicates the evolution of the measured distance. It is clear from the bottom plot that the constrained splines stay within their bounds, which are indicated by the gray region. Observe that the true lower bound is never active due to the conservatism introduced by the linearization. Consequently δ should not be chosen too small. The unconstrained splines clearly exhibit more variation in the distance and seem to follow slow variations of the measured distance. Fig. 8 shows the radial difference Δ of the fitted spline with the true trajectories for both markers. Except for the beginning, the constrained spline (solid line) fits the trajectories more accurately.

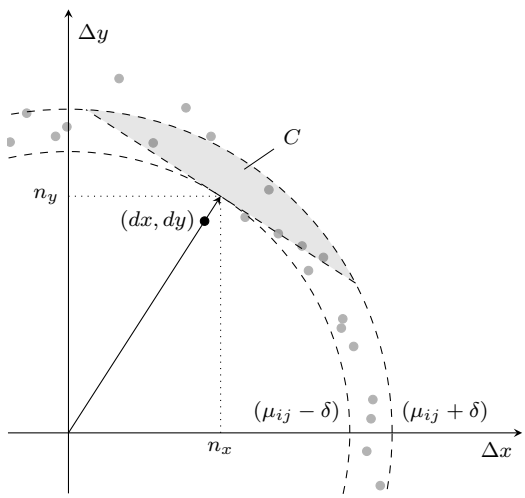

Fig. 5. Linearization of constraint (14) in two dimensions. $(s_{x_i} - s_{x_j}, s_{y_i} - s_{y_j})$ must lie in the ring delimited by the circles with radii $(\mu_{ij} \pm \delta)$. The gray dots indicate measured values.

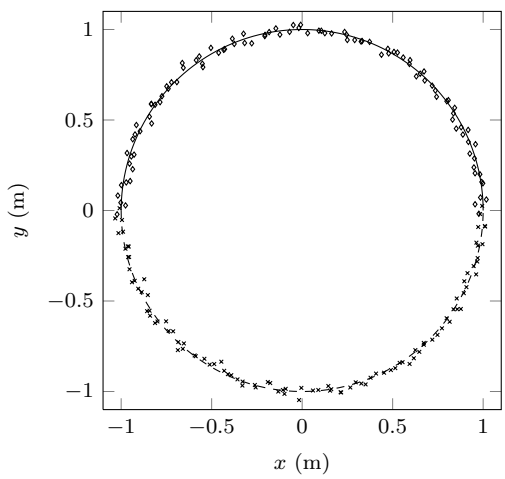

Fig. 6. The simulated marker measurements and true marker trajectories for both markers.

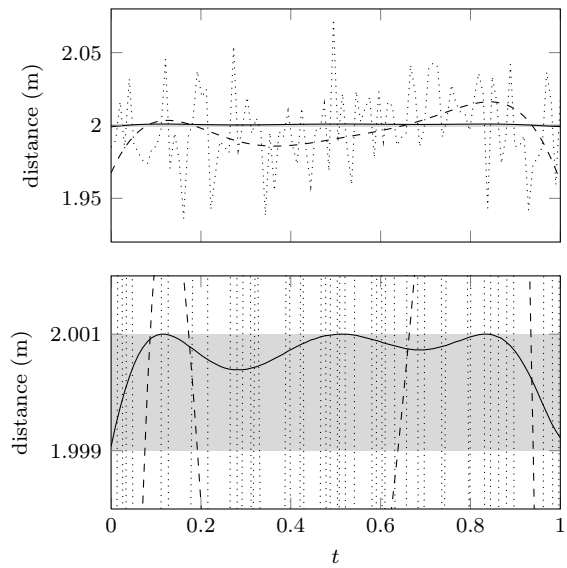

Fig. 7. Resulting distances in function of time for fitted splines with (solid line) and without (dashed line) distance constraints. The dotted line represents the variation of the measured distance. The bottom plot shows the constrained distance in more detail. The distance must lie within the gray region.

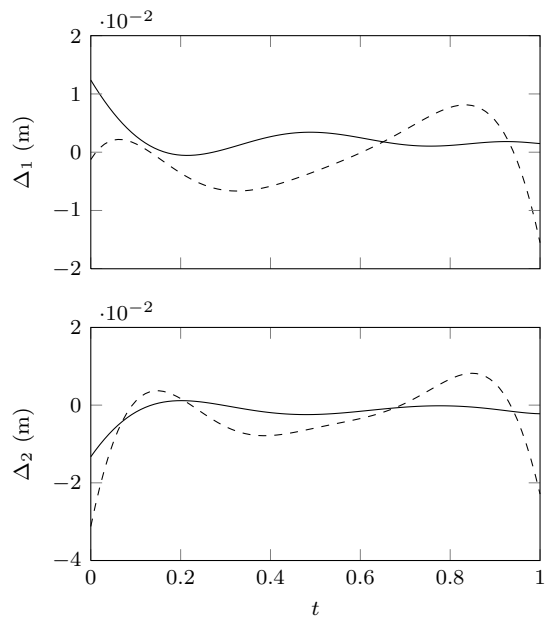

Fig. 8. Radial difference Δ of the fitted splines with the true trajectories for both markers, with (solid line) and without (dashed line) distance constraints.

4. CONCLUSIONS

In this paper we propose a versatile and efficient tool for approximating data with smooth splines. The classical spline approximation problem is reformulated into a convex optimization problem, in which both the number of knots and the knot locations are found automatically and simultaneously. Convex constraints are added easily whereas in the past specialized algorithms were necessary for handling different types of constraints. The numerical results show that, due to the convex reformulation, improved smoothness

values for classical spline fitting problems [Dierckx, 1993] are obtained.

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